

ECO 7938 - Computational and Empirical Methods in Economics 1 Syllabus

Instructor: Gunnar Heins

Email: gheins@ufl.edu

Office Hours: Mondays, 10-11:30 in MAT 320

Class Meeting Times: T & R, 3rd & 4th period (9:35 - 11:30)

Classroom: MAT 108

TA: Pedro de Sousa Almeida (pdesousaalmeida@ufl.edu)

TA Office Hours: Wednesdays, 12-1 in MAT 301A

Course Description

Computational skills have become more and more important in Economics. This course serves as an applied and hands-on introduction to computing and data analysis for graduate students in economics and beyond. The goals of this class are to acquire basic programming skills as well as a working knowledge of two of the most commonly software packages (R and Matlab), to practice working with and analyzing data, and to become familiar with commonly used datasets. This course further provides an introduction to numerical optimization, and UF's computing resources.

Class Modality

This is a fully in-person course. Lectures will not be streamed nor recorded. All exams will take place in person, during our scheduled class times.

Software and Prerequisites

This course will use two programming languages: R during the first half and Matlab during the second one. Prior knowledge of either is not necessary but certainly helpful. Please download and install both R and RStudio (a user interface for R) before the first session. R can be downloaded from <https://www.r-project.org/> and RStudio from <https://www.rstudio.com/>.

Please bring your laptop to each session and have it ready at the beginning of each class. I will for the most part be writing code in R and Matlab during each lecture and you should follow and understand each step by performing it yourself.

Textbooks and Readings

The course will build on multiple sources and does not follow a single textbook, especially for the second part. If you are thinking of getting a textbook on R or Matlab, I would recommend the book by Norman Matloff "The Art of R Programming - A Tour of Statistical Software Design" for the first part. It is useful as an introduction to R but also illustrates quite detailed how R functions compared to other languages. For the basics of Matlab, I recommend the book by Stormy Attaway "Matlab - A Practical Introduction to Programming and Problem Solving".

Exams, Homework, and Grading

Homeworks are an important part of this course and essential in order to learn and practice programming. Especially if you have no or little experience you should use the problem sets as practice for the exams, future courses and ultimately your research. There will be close to weekly homeworks which have to be submitted individually. You can discuss your solutions with others, but do not take over someone else's code.

There will also be three exams. Each exam will be an open-book exam and you will have to use your laptop and use R and Matlab to complete them. It is your responsibility to ensure that your laptop and both programs run properly during the exam. You might also for example want to start important and potentially time-consuming updates before the exam. The final grade will consist of:

Problem Sets	20 %
Participation	5 %
Exam 1	25 %
Exam 2	25 %
Exam 3	25 %
Total	100 %

Make-up assignments and exams will be arranged only for absences that are explicitly covered by the UF Attendance Policy. Whenever possible, you should reach out at least five business days in advance to arrange a make-up assignment or exam. Of course, this will not

always be possible. Unforeseen absences and emergencies occur and can be excused without such advance notice. In most cases, you will be asked to provide evidence or documentation of an absence that is explicitly excused by the UF Attendance Policy. Absences related to religious holidays and worship do not require this documentation.

Professionalism and Honor Code:

Students are bound to not cheat or plagiarize, and are bound by The Honor Pledge which states, “We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: ‘On my honor, I have neither given nor received unauthorized aid in doing this assignment.’”

You should familiarize yourself with the UF Student Honor Code. Cheating and plagiarism are not the only violations of this policy. Making false or misleading statements to procure an improper academic advantage, failing to properly cite quotations, and unauthorized collaboration or consultation of resources are also violations. Importantly, ignorance of a policy is not a valid reason for violating it.

Generative Artificial Intelligence:

The Department of Economics faculty assume that all work that is submitted for grading is written by the student whose name it bears, and that it represents their ideas and work. Accordingly, students are not permitted to use generative AI when completing assignments, quizzes, exams, or other graded work unless their instructor has expressly granted that permission. Unauthorized use of generative AI may constitute cheating and/or plagiarism. Such violations of the UF Student Honor Code will be reported to the UF Dean of Students Office and will be subject to severe sanctions.

Students Requiring Accommodations:

Students with disabilities requesting accommodations should first register with the Disabilities Resource Center (392-8565; <https://disability.ufl.edu/>), providing appropriate documentation. Once registered, students will receive an accommodation letter that can be presented to the instructor when requesting accommodations. Please register at the beginning of the course if seeking accommodations.

Tentative Course Schedule

PART I: PRACTICAL COMPUTING IN R

Week 1

- Course introduction and outline
- Introduction to R and RStudio
- A first R session, Vectors in R
- Loops, Matrices and Lists in R

Week 2

- Installing and using packages
- Working with data: The data.table package
- Importing and saving data
- Summary statistics
- Merging datasets
- Introduction to regression analysis
- Standard errors
- Exporting results

Week 3

- Dummy Variables
- Instrumental Variables, 2SLS
- Bootstrap
- Plots & Data Visualization
- The ggplot2 package

Week 4

- tidyverse
- Simulation
- Applied Empirical Research in Practice
- Further useful R packages, Advantages and Disadvantages of R

Week 5

- tidyverse (cont.)
- Review
- **September 26: Exam 1**

PART II: PRACTICAL COMPUTING IN MATLAB

Week 6

- Introduction to Matlab, Differences to R, a first Matlab session
- Matrices, Loops, and Functions in Matlab

Week 7

- Solving linear equations
- Solving nonlinear equations: Linearization
- Solving nonlinear equations
- Unconstrained Optimization

Week 8

- Unconstrained Optimization (cont.)
- Constrained Optimization
- Speeding up your code, UF's computing resources, HiPerGator

Week 9

- Review
- **October 29: Exam 2**

PART III: AN INTRODUCTION TO APPLIED EMPIRICAL METHODS IN ECONOMICS

Week 10

- Empirical Design
- Causality and Identification
- Diff-in-diff

Week 11

- Instrumental Variables
- Regression Discontinuity Design

Week 12

- Matching
- Panel Data

Week 13

- Quasi-Random Assignment
- Event Studies

Week 14

- **December 3: Exam 3**

佛罗里达大学 - 2024年秋季

ECO 7938 - 计算与实证方法经济学 1

课程大纲

授课教师: Gunnar Heins

邮箱: gheins@ufl.edu

办公时间: 周一, 10-11:30, MAT 320

课程时间: 周二和周四, 第3和第4节课 (9:35 - 11:30)

教室: MAT 108

助教: Pedro de Sousa Almeida (pdesousaalmeida@ufl.edu)

助教办公时间: 周三, 12-1, MAT 301A

课程描述

计算技能在经济学中的重要性日益增加。本课程为经济学及相关领域的研究生提供了一个实用和动手的计算与数据分析入门。课程目标是掌握基本的编程技能, 并熟悉两个最常用的软件包(R和Matlab), 练习处理和分析数据, 并了解常用的数据集。本课程还介绍了数值优化及佛罗里达大学的计算资源。

授课方式

这是一个完全面对面的课程。讲座不会在线直播或录制。所有考试将在规定的课程时间内进行。

软件和先决条件

本课程将使用两种编程语言: 前半部分使用R, 后半部分使用Matlab。不要求事先具备任何知识, 但如果有相关经验会更好。在第一堂课之前, 请下载并安装R和RStudio(R的用户界面)。R可以从<https://www.r-project.org/>下载, RStudio可以从<https://www.rstudio.com/>下载。

请带上您的笔记本电脑参加每节课, 并在每节课开始时做好准备。在每节课上, 我大部分时间将使用R和Matlab编写代码, 您应通过自己动手操作来理解每个步骤。

教材和阅读材料

课程将基于多种资源, 并不遵循单一教材, 尤其是在第二部分。如果您打算购买一本关于R或Matlab的教材, 我推荐Norman Matloff的《The Art of R Programming - A Tour of Statistical Software Design》作为R部分的入门书籍。它不仅对R的入门非常有帮助, 还详细展示了R与其他语言的对比。对于Matlab的基础, 我推荐Stormy Attaway的《Matlab - A Practical Introduction to Programming and Problem Solving》。

考试、作业与评分

作业是本课程的重要组成部分，也是学习和练习编程的关键。尤其是如果您几乎没有相关经验，您应该利用这些习题作为考试、未来课程以及最终研究的练习机会。几乎每周都会有作业需要单独提交。您可以与他人讨论您的解决方案，但不要直接复制别人的代码。

此外，还会有三次考试。每次考试都是开卷考试，您将需要使用笔记本电脑及R和Matlab完成考试。您有责任确保在考试期间您的笔记本电脑及这两个程序正常运行。例如，您可能需要在考试前启动重要且可能耗时的更新。最终成绩的组成如下：

作业：20%

课堂参与：5%

第一次考试：25%

第二次考试：25%

第三次考试：25%

总计：100%

仅在UF考勤政策明确涵盖的缺席情况下，才会安排补考和补作业。尽可能提前五个工作日联系安排补考或补作业。当然，这并不总是可能的。突发缺席和紧急情况的确可能发生，在没有提前通知的情况下，也可以获得批准。在大多数情况下，您将被要求提供UF考勤政策明确认可的缺席证明或文件。与宗教假日和礼拜有关的缺席不需要提供此类文件。

职业道德与荣誉准则

学生必须遵守不作弊或抄袭的规定，并受《荣誉誓言》的约束，该誓言声明：“我们，佛罗里达大学社区的成员，承诺通过遵守荣誉准则来保持我们自己和同伴的最高荣誉和诚信标准。”在佛罗里达大学学生提交的所有作业中，均需要或隐含以下誓言：“在我的荣誉上，我没有在完成此作业时给予或接受未经授权的帮助。”

您应该熟悉佛罗里达大学学生荣誉准则。作弊和抄袭并不是违反此政策的唯一行为。为了获得不正当学术优势而做出虚假或误导性陈述，未能正确引用引文，未经授权的协作或咨询资源等，也都属于违规行为。重要的是，忽视政策并不能成为违规的有效理由。

生成式人工智能

经济学系的教师认为，所有提交评分的作业均由学生本人书写，并代表其思想和工作。因此，除非授课教师明确允许，学生不得在完成作业、测验、考试或其他评分作业时使用生成式AI。未经授权使用生成式AI可能构成作弊和/或抄袭。这类违反UF学生荣誉准则的行为将报告给UF学生事务处，并将受到严厉制裁。

需要提供合理便利的学生

要求提供合理便利的残疾学生应首先向残疾资源中心(392-8565; <https://disability.ufl.edu/>)登记，并提供适当的文件。登记后，学生将收到一封可以在请求合理便利时提交给教师的便利信。如果您需要合理便利，请在课程开始时登记。

课程时间表（暂定）

第一部分：R的实用计算

第1周

课程介绍及大纲

R和RStudio入门

第一次R课程，R中的向量

循环、矩阵和列表在R中的应用

第2周

安装和使用软件包

数据处理：data.table包

导入和保存数据

汇总统计

合并数据集

回归分析入门

标准误差

导出结果

第3周

虚拟变量

工具变量、二阶段最小二乘法(2SLS)

自助法

绘图与数据可视化

ggplot2包

第4周

tidyverse

模拟

实证研究的应用

其他有用的R包，R的优点和缺点

第5周

tidyverse（续）

复习

9月26日：第一次考试

第二部分：Matlab的实用计算

第6周

Matlab入门，与R的区别，第一次Matlab课程

Matlab中的矩阵、循环和函数

第7周

求解线性方程

求解非线性方程：线性化

求解非线性方程

无约束优化

第8周

无约束优化（续）

约束优化

提高代码运行速度，UF的计算资源，HiPerGator

第9周

复习

10月29日：第二次考试

第三部分：应用实证方法在经济学中的介绍

第10周

实证设计

因果关系和识别

差异中的差异（Diff-in-diff）

第11周

工具变量

回归不连续设计

第12周

匹配

面板数据

第13周

准随机分配

事件研究

第14周

12月3日：第三次考试